



Tutorial No. 2

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Problem Statement : Compare various SDLC models and justify the selection of an appropriate model for developing an online shopping system.

Solution :

Theory comparison of SDLC models and selection for online shopping system.

Introduction

Software development life cycle (SDLC) models define structured approaches for developing software systems. Each model has its own advantages and limitations. Selecting an appropriate SDLC model is crucial for the successful development of an online shopping system.

Comparison of various SDLC models :

Waterfall model

Description : A linear and sequential model where each phase must be completed before the next begins.

Advantages :

- Simple and easy to understand
- Well-defined documentation
- Suitable for stable requirements.

Limitations:

Inflexible to requirement changes

Late testing phase

Not suitable for complex and dynamic projects

Incremental model:

Description: The system is developed in small increments, with each increment delivering a working product.

Advantages:

Early delivery of functional components

Flexible to changes.

Easier testing and debugging

Limitations:

Requires good planning

Integration issues may arise.

Prototype model:

Description: A prototype is developed to understand user requirements before actual system development.

Advantages:

Better requirement clarification

Early user feedback.

Reduced risk of requirement mismatch

Limitations:

May lead to poor design if prototype evolves into final product.

Increased development cost

Spiral model

Description: A risk-driven model combining iterative development with systematic risk analysis.

Advantages:

- Effective risk management
- Suitable for large and complex systems
- Continuous user feedback.

Limitations:

- Complex to manage
- High cost
- Requires expertise in risk analysis

Agile model

Description: An iterative and incremental model focusing on flexibility, customer collaboration, and rapid delivery.

Advantages:

- Highly flexible to changes.
- Continuous customer involvement
- Early and frequent delivery.
- Improved quality through continuous testing

Limitations:

- Requires active user involvement
- Difficult to estimate cost and time initially
- Not ideal for very small projects

Selection for appropriate SDLC model for online shopping system.

An online shopping system is dynamic in nature, with frequent changes in user requirements such as payment methods, offers, security

features, and user interface improvements. It also requires continuous testing, fast delivery, and regular updates.

The agile model is the most suitable SDLC model for developing an online shopping system because:

- It supports frequent requirement changes.
- Provides quick delivery of features.
- Encourages continuous customer feedback.
- Ensures better quality through iterative testing.
- Reduces risk through incremental releases.

Conclusion:

Different SDLC models suit different project types. For an online shopping system, which requires flexibility, scalability, and continuous enhancement, the agile SDLC model is the most appropriate choice.